

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ae51ca702899d8ae7.jsonl`  
- SHA-256: `059c3a62d17ab73b66fa571b105ca33b982416a50230d4cd758df15c33b52ee6`  
- Source modified: `2026-06-16T03:23:54+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:22:49.374Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"Frame-wise accuracy (Mean-over-Frames/MoF) does NOT capture over-segmentation ? it can score high even when a single action is fragmented into many discontinuous sub-segments ? whereas segment-based measures, the Edit score (Levenshtein-based) and F1@tau, are the appropriate metrics for detecting over-segmentation."

****Source:**** <https://arxiv.org/pdf/2210.10352> (primary)

****Supporting quote:**** "MoF, as a per-frame calculation, does not capture segment quality; the score can be high even when the segments are fragmented. Dividing an action into many discontinuous sub-segments is referred to as over-segmentation. Segment-based measures like Edit-score [65] and F1-score [52] are instead more suitable measures for evaluating phenomenon like over-segmentation."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:22:52.315Z)

I'll verify this claim about MoF (Mean-over-Frames) accuracy versus segment-based metrics for detecting over-segmentation. Let me check the primary source and search for corroborating/contradicting evidence.

3. assistant (2026-06-16T03:22:54.724Z)

Let me fetch the primary source to verify the supporting quote is accurate.

4. user (2026-06-16T03:22:59.706Z)

Response

****Paper Title & Task:****

"Temporal Action Segmentation: An Analysis of Modern Techniques" ? concerns densely identifying video frames in lengthy videos across multiple action categories.

****Regarding Evaluation Metrics:****

The provided webpage content does not contain any discussion of Mean-over-Frames (MoF) accuracy, frame-wise accuracy, over-segmentation, Edit score, F1-score, or segment-based metrics. The abstract mentions "prevalent evaluation measures" exist but provides no specific details about these metrics or their comparative strengths in capturing over-segmentation issues.

To locate such technical discussions, you would need to access the full PDF paper itself, which this webpage excerpt does not include.

5. user (2026-06-16T03:23:03.792Z)

Web search results for query: "temporal action segmentation MoF frame-wise accuracy does not capture over-segmentation edit score F1 segmental metric"

Links: [{"title": "Temporal Action Segmentation: A Comprehensive Overview", "url": "https://www.emergentmind.com/topics/temporal-action-segmentation"}, {"title": "GitHub - nus-cvml/awesome-temporal-action-segmentation: A curated list of awesome temporal action segmentation resources. · GitHub", "url": "https://github.com/nus-cvml/awesome-temporal-action-segmentation"}, {"title": "Assembly101: A Large-Scale Multi-View Video Dataset for Understanding Procedural Activities", "url": "https://arxiv.org/pdf/2203.14712"}, {"title": "Temporal Action Segmentation: An Analysis of Modern ...", "url": "https://arxiv.org/pdf/2210.10352"}, {"title": "Iterative Contrast-Classify For Semi-supervised Temporal Action Segmentation", "url": "https://ojs.aaai.org/index.php/AAAI/article/download/20124/19883"}, {"title": "Temporal Action Segmentation: An Analysis of Modern Techniques", "url": "https://www.researchgate.net/publication/374971079_Temporal_Action_Segmentation_An_Analysis_of_Modern_Techniques"}, {"title": "Action Segmentation with Mixed Temporal Domain Adaptation", "url": "https://arxiv.org/pdf/2104.07461"}, {"title": "Video action segmentation by mixed temporal domain adaption", "url": "https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11138441"}, {"title": "Point-Supervised Skeleton-Based Human Action Segmentation", "url": "https://arxiv.org/pdf/2603.06201"}]

Based on the web search results, here's what I found about temporal action segmentation metrics and how frame-wise accuracy relates to over-segmentation and other metrics:

Key Findings

Frame-wise accuracy (MoF) does not penalize over-segmentation errors explicitly, which is why segment-wise edit distance (Edit) and F1 scores are also reported.

More specifically, frame-wise accuracy does not account for temporal dependencies of predictions and long action classes have higher impact on this metric than shorter ones, making it unable to reflect over-segmentation errors.

To address these limitations, two alternative metrics are used:

1. ****Edit Score****: The segmental edit score penalizes over-segmentation by measuring the ordering of predicted action segments independent of slight temporal shifts.
2. ****F1 Score****: The segmental F1 score (F1@k) penalizes over-segmentation errors while ignoring minor temporal shifts between predictions and ground truth.

Since temporal segmentation is prone to over-segmentation where an action is fragmented into multiple discontinuous segments, frame-wise accuracy alone cannot adequately reflect this issue, whereas Edit and F1 scores are more suitable for evaluating such phenomena.

These complementary metrics are why the community has adopted this multi-metric evaluation approach for temporal action segmentation tasks.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:23:06.575Z)

The WebFetch couldn't access the PDF body, but the WebSearch strongly corroborates the claim. Let me get the exact quote from the primary PDF and confirm the secondary literature consensus.

7. user (2026-06-16T03:23:12.732Z)

I've reviewed the provided PDF content, which appears to be a heavily compressed/encoded document with embedded images and metadata.

****I cannot locate the specific sentences you're asking about.**** The quoted text regarding:

- "MoF, as a per-frame calculation, does not capture segment quality"
- "Segment-based measures like Edit-score and F1-score are instead more suitable measures for evaluating phenomenon like over-segmentation"

These exact phrases do not appear in the readable portions of the document provided. The content consists primarily of binary-encoded PDF objects, compressed streams, and image data that cannot be reliably extracted as plain text.

To verify these quotes, I would need either:

1. A plaintext version of the source material
2. An uncompressed or OCR'd PDF
3. The original document in a more accessible format

[Binary content (application/pdf, 1.8MB) also saved to C:\Users\avidu\claudes\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579999760-41alwg.pdf]

8. user (2026-06-16T03:23:16.664Z)

Web search results for query: "MS-TCN segmental F1 edit score over-segmentation frame-wise accuracy not penalize temporal action segmentation Lea Farha"

Links: [{"title": "[PDF] MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/MS-TCN:-Multi-Stage-Temporal-Convolutional-Network-Farha-Gall/b4673e744d0ded47fe6df3b6314f79a41359578b"}, {"title": "1 MS-TCN++: Multi-Stage Temporal Convolutional Network for Action Segmentation", "url": "https://yunliu.github.io/papers/(TPAMI'2020)MS-TCN++%20-%20Multi-Stage%20Temporal%20Convolutional%20Network%20for%20Action%20Segmentation.pdf"}, {"title": "Video action segmentation by mixed temporal domain adaption", "url": "https://image-pubs.uspto.gov/dirsearch-public/print/downloadPdf/11138441"}, {"title": "(PDF) MS-TCN++: Multi-Stage Temporal Convolutional Network for Action Segmentation", "url": "https://www.researchgate.net/publication/344127554_MS-TCN_Multi-Stage_Temporal_Convolutional_Network_for_Action_Segmentation"}, {"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation", "url": "https://openaccess.thecvf.com/content_CVPR_2019/papers/Abu_Farha_MS-

```
TCN_Multi-Stage_Temporal_Convolutional_Network_for_Action_Segmentation_CVPR_2019_paper.pdf"}, {"title": "Temporal Action Segmentation: A Comprehensive Overview", "url": "https://www.emergentmind.com/topics/temporal-action-segmentation"}, {"title": "Temporal Action Segmentation: An Analysis of Modern ...", "url": "https://arxiv.org/pdf/2210.10352"}, {"title": "Assembly101: A Large-Scale Multi-View Video Dataset for Understanding Procedural Activities", "url": "https://arxiv.org/pdf/2203.14712"}, {"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation", "url": "https://www.researchgate.net/publication/338511817_MS-TCN_Multi-Stage_Temporal_Convolutional_Network_for_Action_Segmentation"}, {"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation", "url": "https://arxiv.org/pdf/1903.01945"}]
```

I found relevant search results about MS-TCN, temporal action segmentation, and evaluation metrics. Here are the key findings:

MS-TCN and Evaluation Metrics

MS-TCN is a temporal convolutional network that employs multi-stage 1D convolutions with dilations to aggregate both local and global temporal context, developed by Farha et al.

Evaluation Metrics

The main evaluation metrics used for temporal action segmentation include:

- Frame-wise Accuracy**: Frame-wise accuracy (Acc) is one of the standard metrics, along with segmental edit distance and segmental F1 score at overlapping thresholds of 10%, 25% and 50% (F1@{10, 25, 50})
- Segmental F1 and Edit Score**: Segmental F1@IoU is the F1 score over predicted vs. true segments, considering only those with temporal IoU exceeding thresholds (typically 10%, 25%, 50%). These metrics explicitly penalize over-segmentation and boundary errors
- Over-segmentation Handling**: Over-segmentation is mitigated through per-stage refinement and truncated mean-squared smoothing on the log-probabilities

Frame-wise Accuracy and Over-segmentation

Frame-wise accuracy (MoF) does not penalize over-segmentation errors explicitly, while segment-wise edit distance (Edit) and F1 scores at overlapping thresholds do penalize these errors

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-16T03:23:20.236Z)

The PDF binary couldn't be parsed by WebFetch, but I have the file saved locally. Let me extract the text directly to confirm the verbatim quote, and the secondary literature is already overwhelmingly corroborating.

10. user (2026-06-16T03:23:24.446Z)

```
-rw-r--r-- 1 avidu 197609 1488854 Jun 16 08:48 webfetch-1781579908809-uip056.pdf
-rw-r--r-- 1 avidu 197609 1444663 Jun 16 08:48 webfetch-1781579910495-xf7r2b.pdf
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```

Traceback (most recent call last):

11. assistant (2026-06-16T03:23:27.197Z)

The 2210.10352 PDF is the 1.8MB one (webfetch-...-41alwg.pdf). Let me extract its text. No PDF library is available, so I'll try a different approach.

12. user (2026-06-16T03:23:34.371Z)

[notice] A new release of pip is available: 26.0.1 -> 26.1.2

[notice] To update, run: python.exe -m pip install --upgrade pip

== per-frame ==

have large qualitative differences, suggesting that class-balanced metrics may be more appropriate though this is currently not adopted in the literature. Furthermore, MoF, as a per-frame calculation, does not capture segment quality; the score can be high even when the segments are fragmented. Dividing an action into many discontinuous sub-segments is referred to as over-segmentation. Segment-based measures

== over-segment ==

lulation, does not capture segment quality; the score can be high even when the segments are fragmented. Dividing an action into many discontinuous sub-segments is referred to as over-segmentation. Segment-based measures like Edit-score [65] and F1-score [52] are instead more suitable measures for evaluating phenomenon like over-segmentation. Segment-Based Measures. The Edit Score [65] quantifies the similarity

== oversegment ==

follow frame-wise prediction (Eq. (2)). We recommend a greater emphasis on solving the task at the segment level and anticipate that it can greatly reduce the current issues with oversegmentation. Forms of Supervision. Procedural video sequences feature enormous temporal redundancy in the supervisory signals due to the significant similarity between successive video frames of the same motion. Such redundancy

== Edit score ==

times the number of frames compared to the least frequent class. On the JOURNAL OF LATEX CLASS FILES, VOL. 14, NO. 8, AUGUST 2015 5 GT: Pred: Diff: (a) MoF GT: Pred: Edit Dist.: (b) Edit Score GT: Pred: Det.: (c) F1 Score Fig. 5: Evaluation metric visualization. (a) MoF estimates how accurate frame wise predictions are. (b) Edit score tolerates small boundary shifts, as long as the sequence order is correct.

== segment quality ==

gesting that class-balanced metrics may be more appropriate though this is currently not adopted in the literature. Furthermore, MoF, as a per-frame calculation, does not capture segment quality; the score can be high even when the segments are fragmented. Dividing an action into many discontinuous sub-segments is referred to as over-segmentation. Segment-based measures like Edit-score [65] and F1-score [52] are

== MoF ==

uent action class has only 6 times the number of frames compared to the least frequent class. On the JOURNAL OF LATEX CLASS FILES, VOL. 14, NO. 8, AUGUST 2015 5 GT: Pred: Diff: (a) MoF GT: Pred: Edit Dist.: (b) Edit Score GT: Pred: Det.: (c) F1 Score Fig. 5: Evaluation metric visualization. (a) MoF estimates how accurate frame wise predictions are. (b) Edit score tolerates small boundary shifts, as long

13. assistant (2026-06-16T03:23:44.975Z)

The verbatim quote is confirmed in the primary source. The extracted text matches the supporting quote exactly:

> "Furthermore, MoF, as a per-frame calculation, does not capture segment quality; the score can be high even when the segments are fragmented. Dividing an action into many discontinuous sub-segments is referred to as over-segmentation. Segment-based measures like Edit-score [65] and F1-score [52] are instead more suitable measures for evaluating phenomenon like over-segmentation."

The source is "Temporal Action Segmentation: An Analysis of Modern Techniques" (arXiv:2210.10352), a peer-reviewed survey (later published in IEEE TPAMI). The claim is well-established in the field. Let me run the verification across the checklist:

1. **Quote support**: The claim is an accurate paraphrase of the quote ? not an overreach. MoF = per-frame, high even when fragmented = over-segmentation not captured; Edit + F1 = segment-based + suitable for over-segmentation. Exact match.

2. **Contradicting evidence**: None found. The opposite ? multiple independent sources (MS-TCN/Farha & Gall, the segmentation surveys) corroborate that frame-wise accuracy does not penalize over-segmentation, which is the entire reason segmental F1@k and the edit score were introduced.

3. **Source quality**: Primary, peer-reviewed survey (TPAMI-grade). Strong for a non-extraordinary, widely-accepted methodological claim.

4. **Outdated**: 2022/2023 survey; the underlying point dates to Lea et al. 2016 / MS-TCN 2019 and remains current standard practice. Not outdated.

5. **Marketing**: No ? it is a methodological observation in an academic survey, not a benchmark or press release.

One minor nuance worth flagging (not a refutation): the claim says Edit/F1 are "the appropriate metrics for detecting over-segmentation." More precisely they *penalize* it; the most direct over-segmentation-specific diagnostic is the segment-count-ratio. But this is a refinement, not a contradiction ? the claim as stated is fully supported.

14. user (2026-06-16T03:23:51.685Z)

Structured output provided successfully

15. assistant (2026-06-16T03:23:54.256Z)

Claim verified as supported. The verbatim quote was confirmed in the primary source (arXiv:2210.10352) by extracting the PDF text directly, and the methodological point is independently corroborated by MS-TCN and the broader temporal action segmentation literature. refuted=false, high confidence.